When and Why Does Nonresponse Occur? Comparing the Determinants of Initial Unit Nonresponse and Panel Attrition

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Abstract

Though unit nonresponse threatens data quality in both cross-sectional and panel surveys, little is understood about how initial nonresponse and later panel attrition may be theoretically or empirically distinct phenomena. This study advances current knowledge of the determinants of both unit nonresponse and panel attrition within the context of the U.S. Census Bureau's Survey of Income and Program Participation (SIPP) panel survey, which I link with high-quality federal administrative records, paradata, and geographic data. By exploiting the SIPP's interpenetrated sampling design and relying on cross-classified random effects modeling, this study quantifies the relative effects of sample household, interviewer, and place characteristics on baseline nonresponse and later attrition, addressing a critical gap in the literature. Given the reliance on successful record linkages between survey sample households and federal administrative data in the nonresponse research, this study also undertakes an explicitly spatial analysis of the place-based characteristics associated with successful record linkages in the U.S.

Keyword*:* Unit Nonresponse, Panel Attrition, Refusal, Noncontact, Interviewer Effects, Area Effects

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INTRODUCTION

As survey unit nonresponse in both cross-sectional and panel surveys has continued to grow in recent decades (Williams and Brick 2018), researchers have sought to identify the factors contributing to this trend and the ways in which they weaken data quality (Brick and Tourangeau 2017). This study builds upon this research in several important ways, first by comparing the correlates of initial unit nonresponse and that of panel attrition, which has received very little attention in the extant literature, and second, by distinguishing between the separate effects of interviewers and places on survey response behaviors.

One of the greatest challenges to studying unit nonresponse in the United States is the dearth of detailed individual, household, and neighborhood data on nonresponders. Such data is largely inaccessible or unavailable. For this reason, among others, no study to date has been able to adequately compare the determinants of unit nonresponse with that of panel attrition, particularly when broken down by refusal and noncontact, leading to many assumptions about how the determinants of one type of nonresponse are related to the other. Fortunately, recent efforts by the U.S. Census Bureau to link federal survey data with high-quality administrative records (Wagner and Layne 2014) offer one potential path forward. In this study, I exploit new administrative data linkages with both survey nonresponders and panel attriters in the Survey of Income and Program Participation (SIPP), a face-to-face federal panel survey. Currently, a wealth of evidence points to the individual and household characteristics associated with lower response propensities in cross-sectional and panel surveys (e.g., Johnson et al. 2002; Frankel and Hillygus 2013), but the determinants that extend beyond the individual are relatively less clear. One prominent line of research investigates the role of interviewers in securing cooperation, with evidence to suggest that certain interviewers are better able to elicit participation than others (Castiglioni et al. 2008; Frankel and Hillygus 2013; Schaeffer et al. 2010; West and Blom 2017). And, as separate research has demonstrated, certain features of the social environment—such as urbanicity and crime, among others—are also linked with nonresponse (Demaio 1980; Brick and Williams 2013; Goyder et al. 1992; Groves and Couper 1998; Johnson et al. 2006; Smith and Son 2010).

This presents yet another challenge in the research on unit nonresponse, as interviewers are typically confined to a specific location in face-to-face interviews, confounding the effects of the interviewer and place. Surveys with interpenetrated sample designs, whereby interviewers

are not perfectly nested within places, and places are not perfectly nested within interviewers, serve as one potential solution to this estimation problem, first proposed by Campanelli and O'Muircheartaigh (1999). The SIPP is among the few surveys that employs interpenetrated interviewer assignment, allowing data users to distinguish between the features of the interviewer versus the social environment that contribute to nonresponse. This study therefore makes an additional contribution to the literature by explicitly accounting for interviewer interpenetration, allowing me to disentangle the respondent, interviewer, and place-based sources of variation in nonresponse and attrition.

To do so, I compare respondents who participated in the first two survey waves with nonresponders (in wave 1) and attriters (in wave 2) in the 2014 SIPP panel survey. Given important differences in the predictors of nonresponse depending on whether interviewers are able to establish contact with sample members (Laurie, Smith, and Scott 1999), the present study differentiates between noncontact and refusals, thereby estimating the predictors of four major nonresponse outcomes—nonresponse due to noncontact, nonresponse due to refusal, attrition due to noncontact, and attrition due to refusal. By specifying interviewer and county-level groupings as "crossed" random effects, I estimate the relative proportion of variance attributable to each term across the four outcomes, and make direct comparisons of both the fixed and random effects that contribute to each response outcome. Because these analyses only include SIPP sampling units successfully linked with federal administrative records, I augment these findings with a separate analysis of the place-based characteristics of "unlinkable" sampling units that accounts for the significant degree of spatial autocorrelation in place-based linkage rates.

BACKGROUND

Unit nonresponse, which occurs when a sampling unit fails to participate in a survey, can occur in cross-sectional surveys, which are administered at only one point in time, and in panel surveys, which interview the same set of respondents at different points in time. In much of the survey methodology literature, *nonresponders*, or sampling units that never participated in a given survey, and *attriters*—sampling units that discontinue their participation in a panel survey after cooperating in the initial survey wave—are often viewed with the same lens, understood using the same theories of survey participation. The prevailing conceptual model (Groves et al. 1992; Groves and Couper 1998) considers the interplay of characteristics of the sample householder (e.g., ethnoracial identity), social environment (e.g., urbanicity), interviewer (e.g., level of experience), and survey design (e.g., incentives). Together, these forces are thought to coalesce during the actual survey request, shaping the sample member's brief encounter with the interviewer and, ultimately, their decision to participate. Several additional factors presumably inform their decision, including the perceived importance (or "leverage") and salience of different attributes of the survey design, including survey participation incentives and the legitimacy of the sponsoring research institution (Groves et al. 2000), as well as past experiences and perceived rewards associated with survey participation (Goyder et al. 2006).

Importantly, however, these theories have been predominantly tested using cross-sectional surveys, tempering their potential application to panel surveys (Laurie 2008; Kocar and Lavrakas 2023). This is not surprising given the broader lack of empirical evidence distinguishing the characteristics of nonresponders from that of attriters, with very few exceptions (e.g., see Pyy-Martikainen and Rendtel 2008; Fuchs 2012). In one of the only direct comparisons of the correlates of initial nonresponse versus panel attrition, Lugtig et al. (2014) found significant differences in age, household composition, household size, and urbanicity

between nonresponders and attriters in an online panel survey. Because participation in web-based surveys is partially driven by the sample member's access to or comfort with the Internet and email, Lugtig and colleagues posit that there are likely more similarities between the predictors of nonresponse and panel attrition in samples recruited offline than online, generating important questions about the extent to which these predictors overlap in interviewer-administered surveys[1].

In the absence of this type of research, separate studies on the determinants of unit nonresponse and panel attrition can be used to deduce common patterns. As alluded to previously, there are major differences in the availability of data on nonresponders versus attriters. Whereas individual or household characteristics of attriters can be collected from a prior survey in which they participated, characteristics of nonresponders in the general population can only be sourced from federal administrative records, which are highly inaccessible in the U.S. In the absence of such records, researchers have relied more heavily on place-based data to examine neighborhood-level differences between nonresponders and respondents in surveys based on area or address frames (Ellis et al. 1970; Groves and Couper 1998; Lugtig et al. 2014). As a result, research on individual or household-level determinants is much more developed in the panel attrition literature, whereas the effect of neighborhood characteristics, in which neighborhood characteristics are sometimes treated as proxies for the characteristics of the nonresponder, is much more developed in the nonresponse literature. In the following sections, I review existing research on the sociodemographic characteristics of nonresponders and attriters, and how they vary depending on the type of nonresponse (i.e., noncontact versus refusal); I also provide an overview of the research on interviewer and area effects.

[1] Existing comparisons of nonresponders and attriters in the literature (i.e., Pyy-Martikainen and Rendtel 2008; Fuchs 2012; Lugtig et al. 2014) exclusively rely on European survey data.

Nonresponders and Attriters

The nonresponse research with access to administrative records reveals interesting similarities—and differences—with the panel attrition research, though this literature is quite limited. For instance, in their review of the literature on the racial determinants of unit nonresponse versus panel attrition, Johnson et al. (2002) found little evidence of racial differences in nonresponse[2] but significant differences in panel attrition, such that respondents of color were more likely to drop out of panel surveys than their White counterparts, a finding that has been echoed by later works on the determinants of panel attrition (e.g., Radler and Ryff 2010). Additionally, beyond studies included in the 2002 review by Johnson et al., evidence on the relationship between race/ethnicity and unit nonresponse is more mixed. Treat (2017), for example, finds that Black sample members are less likely to respond than other racial/ethnic groups in the SIPP, but a review of the relevant literature by Sykes et al. (2010) finds little evidence of significant racial or ethnic differences in unit nonresponse.

Bee et al. (2015) find no income disparities between respondents and nonrespondents in the Current Population Study (CPS), but these findings may be undermined by the fact that lower-income sample members are less likely to be linked with administrative records in federal surveys (Bond et al. 2014), a topic I discuss in greater detail later on. The relationship between income and attrition is also complicated. Fitzgerald et al. (1998), for instance, find that respondents with unstable earnings are more likely to attrite, whereas Gray et al. (1996) find that low-income respondents are more likely to attrite.

[2] Of the six studies on cross-sectional nonresponse Johnson et al. (2002) review, only three of them examine the general U.S. population, and one of these studies limits analysis to nonresponders due to refusal (rather than noncontact).

Further complicating these findings is the distinction between refusal and noncontact in face-to-face surveys (Lin and Schaeffer 1995). Attriters who refuse to continue participation often represent a substantively different set of people than attriters with whom contact cannot be reestablished (Laurie, Smith, and Scott 1999). Respondents who move from one residence to another are often the most difficult to contact (Trappmann et al. 2015), and geographically mobile respondents tend to be younger and more educated than those who maintain stable home addresses (Lepkowski and Couper 2002). Respondents who experience a major life event (e.g., marriage, divorce, job change) are more likely to be geographically mobile, and they are also harder to track when that life event results in a name change alongside the move (Lepkowski and Couper 2002). At the same time, a substantial portion of refusals among attriters are "situational," Burton et al. (2006) find, such that "a particular event or stressful situation leads to a refusal one year but is essentially temporary" (p. 463). In contrast, among initial nonresponders, there is evidence to suggest that physical barriers to access are significant sources of noncontact (Groves and Couper 1998); these include "locked central entrances to apartment buildings, walled subdivisions, locked gates, intercoms" (Dillman et al. 2002: p. 7), which can appear across households at different levels of the income distribution (Groves and Couper 1998). Even when the sample unit is accessible to the interviewer, however, interviewers can still fail to make contact when no one is home, the likelihood of which varies across subgroups of the population; households with elderly people and young children tend to be home more frequently, whereas employed, single-person households are less likely to be home (Groves and Couper 1998).

Interviewers

Interviewer characteristics have also been shown to influence nonresponse and attrition (Couper and Groves 1992; Castiglioni et al. 2008; Schaeffer et al. 2010; Brunton-Smith et al. 2012; Frankel and Hillygus 2013; West and Blom 2017; Ackermann-Piek et al. 2020). Surveys with more experienced and better trained interviewers tend to garner greater sample retention in panel surveys (Castiglioni et al. 2008; Frankel and Hillygus 2013) and greater cooperation overall (Couper and Groves 1992; West and Blom 2017; Ackermann-Piek et al. 2020). Interviewers that are better able to recruit reluctant respondents (i.e., convert potential refusals) also appear to be relatively successful at reducing noncontact, as they tend to work harder to track and locate respondents (Schaeffer et al. 2010; Brunton-Smith et al. 2012). Depending on their residential proximity to sample households, face-to-face interviewers may also possess more "local knowledge" to aid them in obtaining cooperation (Groves and Couper 1998). According to several studies, interviewer continuity is an important predictor of sample retention in panel surveys, such that respondents are more likely to continue participation in a panel survey if they are paired with the same interviewer across waves (Laurie et al. 1999; Lepkowski and Couper 2002; Hill and Willis 2001). As Lynn et al. (2014) point out, however, there is little causal evidence linking interviewer continuity with attrition, with evidence from a series of survey experiments rejecting the relationship entirely (Campanelli and O'Muircheartaigh 1999, 2002).

Given these mixed findings[3], an important point to consider when evaluating the effect of interviewer continuity is the extent to which respondents are nonrandomly assigned to the same interviewer across waves in a given panel survey; in the Health and Retirement Study (HRS), for instance, respondents who have a positive survey experience with an interviewer are more likely

[3] Campanelli and O'Muircheartaigh (2002) argue that accounting for *interviewer* attrition in panel surveys would significantly attenuate the effect of interviewer continuity in other studies.

to be paired with that same interviewer in subsequent interviews (Hill and Willis 2001), a practice widely used in many panel surveys (Campanelli and O'Muircheartaigh 2002). In a separate survey experiment in which respondents were, in fact, randomly assigned to an interviewer across waves, Lynn et al. (2014) found that the effect of interviewer continuity was contingent on respondents' age, such that continuity only affected response propensities among respondents below the age of sixty.

Another challenge is the potential change in the effect of interviewer continuity across survey waves. In the CPS, for instance, Rope (1993) demonstrates that interviewer continuity has a smaller effect on attrition toward the end of a panel survey than it does during the initial survey waves, with the effect of interviewer continuity dissipating after the household leaves the sample rotation for eight months. Finally, although certain interviewer sociodemographic characteristics have been shown to influence response propensities (West and Blom 2017), the effects tend to be small, which may be partly attributable to the general homogeneity among survey interviewers along characteristics like race, gender, and age (Schaeffer et al. 2010). Still, much of the research on interviewer sociodemographic characteristics is restricted to examining unit nonresponse in cross-sectional surveys or in baseline interviews of panel surveys; little is known about the degree to which such interviewer characteristics influence respondents' decisions to participate in subsequent waves of panel survey, with some exceptions (e.g., see Campanelli and O'Muircheartaigh 1999; Lynn et al. 2014).

Interviewer versus Area Effects

Disentangling interviewer effects from "area effects" presents another important challenge in the research on interviewers and unit nonresponse. In face-to-face interviews,

interviewers are often assigned to a specific geographic area, and respondents in certain areas may be more motivated to respond than others. As a separate line of research has demonstrated, there are a number of features of the social environment that are linked with unit nonresponse (House and Wolf 1978; Demaio 1980; Goyder et al. 1992; Groves and Couper 1998; Campanelli and O'Muircheartaigh 1999; Johnson et al. 2006; Smith and Kim 2009; Smith and Son 2010; Brick and Williams 2013; Bell et al. 2019; Okubo and Yokoyama 2020). The social environment, in this context, is a facet of the "survey-taking climate" (Lyberg and Dean 1992; Lyberg and Lyberg 1991), which broadly refers to the "public's general willingness to cooperate in surveys and [the] societal and organizational factors that influence this willingness," (Loosveldt and Joye 2016: 69).

Within the U.S., markers of concentrated neighborhood disadvantage—such as poverty (Johnson et al. 2006), unemployment, lower educational attainment, and greater proportion of renter-occupied households (Bell et al. 2019)—appear to positively predict unit nonresponse. At the same time, there is also evidence to suggest that individuals in areas with concentrated affluence are also less likely to participate in surveys through both refusal and noncontact (Johnson et al. 2006). There is substantial evidence to suggest that urbanicity is negatively associated with survey participation (Demaio 1980; Goyder et al. 1992; Groves and Couper 1998; Johnson et al. 2006). Though considerably less research has been devoted to identifying the place-based determinants of panel attrition, there are a few notable studies. Smith and Son (2010) found that residents of large central cities were less likely to continue participation in follow-up GSS panels, and Gray et al. (1996) found greater attrition among urban British residents. In their comparison of nonresponders and attriters from the same online panel survey, Lugtig et al. (2014) found that nonresponders lived in significantly more urban areas than

attriters, who lived in significantly more urban areas than respondents who participated in all survey waves within the panel.

Given the significance of place, distinguishing between the features of the interviewer versus the "survey-taking climate" that contribute to nonresponse becomes a difficult task. In their 1999 paper, Campanelli and O'Muircheartaigh offered one solution to this particular estimation problem, which necessitates an interpenetrated sample design. Now considered the gold standard in this line of research, a fully interpenetrated sample design randomly assigns interviewers to households regardless of location. In practice, however, this is prohibitively expensive to implement given travel costs associated with moving interviewers from location to location, resulting in the prominence of partial interpenetration, in which interviewers are not fully nested within places, and places are not fully nested within interviewers. With an interpenetrated sample design, a cross-classified multilevel model with random terms for the interviewer and place can be used to distinguish between the two sources of variation.

Given the high costs associated with even partial interpenetration, however, studies that exploit this sampling design strategy to isolate the effects of interviewers from places are quite rare and, furthermore, predict a disparate set of outcomes, including panel attrition (Berete et al. 2019; Campanelli and O'Muircheartaigh 1999), unit nonresponse (Durrant et al. 2010; Ackermann-Piek et al. 2020), and item response patterns (Leone et al. 2021). In the absence of interpenetrated sample designs, scholars have also taken other creative approaches to studying interviewer versus area effects (e.g., see work by Haunberger 2010; Mierzwa et al. 2002; Schnell and Kreuter 2005[4]). Across these studies, the relative effect of the interviewer varies depending

[4] Schnell and Kreuter (2005) make use of a mixed-mode design, whereby respondents are randomly assigned to mail surveys or face-to-face interviewing, and interviewers are hierarchically nested within places rather than interpenetrated assignment. Perceptions of neighborhood crime differed across mode, even among

on the specific outcome. Campanelli and O'Muircheartaigh (1999), for instance, find that in cases of panel attrition due to refusal, interviewer effects supersede place effects, but that, perhaps unsurprisingly, panel attrition due to noncontact is driven more by place effects than by interviewer effects. In contrast, however, Ackermann-Piek et al. (2020) estimate greater interviewer effects on a sample member's propensity to be contacted than on their propensity to cooperate, though the authors analyze nonresponse in cross-sectional rather than panel surveys, unlike Campanelli and O'Muircheartaigh (1999). In their analysis of unit nonresponse due to refusal, Durrant et al. (2010) find that both interviewer and place effects are significant when the household characteristics of sampling units are not accounted for, but that the significance of the place goes away after the inclusion of household-level variables, generating additional questions about the impact of places on nonresponse due to noncontact. There is evidence to suggest that the specific features of the social environment that are accounted for may also play a pivotal role; for instance, in their cross-national analysis, Leone et al. (2021) find that the effect of interviewers on the number of reported abortions is significantly greater than that of places, but only after controlling for region and urbanicity. It is unclear whether these findings would necessarily extend to work on unit nonresponse or even on responses to other types of survey questions, as respondents may be more apt to rely on "explicit or implicit" help from interviewers when items are sensitive, non-factual, or open (Schnell and Kreuter 2005).

The extant research faces other important limitations. First, studies on interviewer versus area effects seldom include explicit discussions of place or geographic scale. In most of this work, areas remain undefined, instead referred to as "primary sampling units" (Campanelli and O'Muircheartaigh 1999; Durrant et al. 2010), "clusters" (Leone et al. 2021), or even simply as

respondents within the same place, demonstrating the relative importance of interviewers over places in shaping responses.

"areas" (Haunberger 2010), raising questions about the underlying mechanisms that contribute to their effect. If a sample members' propensity to respond is indeed partially shaped by the social environment, what features of the environment are important, and at what scale do they operate? Lyberg and Lyberg do not provide a precise definition of the survey-taking climate in their seminal 1991 paper, leaving much open to interpretation. As a result, some scholars have characterized the survey climate as a reflection of larger societal trends in the proliferation of surveys and survey requests that contribute to "population-level attitudes and beliefs about survey research" (Yan and Datta 2015) and the "general feeling that people are asked to participate in surveys too often" (Loosveldt and Joye 2016). At the local level, scholars point to the importance of urbanicity (and, by extension, poverty and crime), which are thought to contribute to residents' exposure to social disorder and reduced social cohesion and therefore shape survey cooperation (Groves and Couper 1998; House and Wolf 1978; Johnson et al. 2006; Loosveldt and Carton 2002). Others instead investigate local area effects because of their presumed homogeneity (e.g., Ellis et al. 1970; Lugtig et al. 2014). As Schnell and Kreuter (2005) put it, "respondents who live in the same geographical areas… are often more similar to each other than respondents selected at random from the population as a whole," and that, similarly, "people who live in small spatial clusters share similar attitudes because of the similar circumstances in which they live." The proposition here is that neighborhoods simply serve as a proxy for individual or household characteristics, which are thought to be the ultimate determinant of response propensities. Though it is not explicitly stated, the underlying assumption is that the scale at which places are *most* sociodemographically homogenous—i.e., the smallest scale of geographic aggregation—captures the greatest variation in survey outcomes.

STUDY DESIGN

Current Study

By separating unit nonresponse into four categories broken down by type (noncontact versus refusal) and survey wave (baseline nonresponse versus panel attrition), the present study makes direct comparisons of the distinct factors that contribute to each outcome, with a particular focus on comparisons between continuous respondents, initial nonresponders, and panel attriters[5]. I predict that characteristics of the sample household, social environment, and data collection process will differentially influence unit nonresponse depending on *when* it occurs and *why*. Furthermore, by accounting for the distinct roles of interviewers and places in shaping outcomes, I also compare the extent to which each of these groupings (i.e., interviewers and places) differentially explain variation in the precise category of nonresponse. The extant research has not yet distinguished between interviewer and area effects in the U.S., where response propensities may be geographically patterned in distinct ways[6]. I therefore examine interviewer versus area effects among a particularly large sample of Americans, specifically, and across both nonresponders and attriters from the same panel survey. Importantly, my analysis comprises descriptive comparisons across five groups of sample members: nonresponders due to noncontact in wave 1, nonresponders due to refusal in wave 1, attriters due to noncontact in wave 2, attriters due to refusal in wave 2, and "continuous" respondents, who participated in both waves. As a result, I do not address the temporal process that contributes to response outcomes in

[5] It is not entirely clear whether nonresponders in the first wave of a panel survey are fully comparable with nonresponders in a cross-sectional survey, as "the awareness of the subsequent waves might cause slightly more refusals to participate in a panel study than to participate in a cross-sectional one" (Junes 2012: p. 6).
[6] Previous studies have investigated interviewer versus area effects in separate European states (Campanelli and O'Muircheartaigh 1999; Schnell and Kreuter 2005; Durrant et al. 2010; Berete et al. 2019; Ackermann-Piek et al. 2020), and across several countries in sub-Saharan Africa, Europe, Asia, and Latin America (Leone et al. 2021).

the second wave (i.e., I do not account for wave 1 response propensities of respondents included in the wave 2 analysis.) Through the use of high-quality, federal administrative records and linkages with third-party data, I account for the household and place-based characteristics of nonresponders and attriters, among other covariates, to better understand the role of interviewers and places in shaping response behavior.

However, even when information on nonresponders is accessible, the quality and/or representativeness of that data is limited. For instance, in Lugtig et al.'s 2014 study comparing the determinants of initial nonresponse with that of panel attrition, the authors' analysis is limited to nonresponders who actually completed an initial recruitment interview; nonresponders who refused to participate in this recruitment interview or who could not be contacted (i.e., more than half of all nonresponders) were therefore omitted from their analysis. To obtain administrative records for the complete sample of nonresponders, studies typically rely on a specialized sampling frame (e.g., university students; see Patrick et al. 2013) limiting the extent to which their findings can be generalized to the broader population.

Federal administrative records, which perhaps offer the highest quality information on nonresponders, also face their own set of problems. To link such records with sampling units, the Census Bureau's Person Identification Validation System (PVS) assigns a Protected Identification Key (PIK)—i.e., a unique person identifier—to each individual within a household sampling frame for whom a unique record in the reference files can be identified (Wagner and Layne 2014). Individuals who are assigned a PIK can be linked across all Census data files that have been processed using the PVS (Wagner and Layne 2014). In practice, however, not everyone can be assigned a PIK due to insufficient information, generating nonrandom variation in record linkages across different subsets of the population. Using the Census Bureau's

American Community Survey (ACS) as an example, Bond et al. (2014) find that many marginalized groups, including people of color, immigrants, recent movers, as well as low-income and unemployed individuals are less likely to be assigned a PIK, and therefore less likely to be linked with administrative records, with potential implications for existing findings on sociodemographic differences between nonresponders and respondents in the general population. This nonrandom missingness in federal records is not, by any means, a new problem; in *Looking for the Last Percent* (1984), Choldin concludes his assessment of the Decennial Census undercount in the United States with a simple statement, writing that "people in certain situations have good reasons to avoid contact with the government" (p. 230). Perhaps the most obvious example of this is the fear of deportation among undocumented immigrants, but there is also the unintended impact of laws related to public housing, welfare, zoning, and occupancy; for example, "if two families are doubled up in public housing unit, in violation of their lease, they may attempt to conceal some of the occupants" (Choldin 1984:231).

As such, my analysis of the determinants of unit nonresponse and panel attrition only includes members of the SIPP sampling frame who have been assigned a PIK—i.e., 91% of all sample members, and 78% of initial nonresponders—potentially biasing my estimates. To better understand the characteristics of this omitted sample, the present study also examines the degree of place-based variation in administrative linkages. Given the presence of spatial clustering in Census response rates (Griffith and Chun 2015) and among certain sociodemographic groups in the U.S., there is good reason to question whether sample units in certain neighborhoods are more identifiable from a data perspective, with enormous implications for research on survey nonresponse. I take an explicitly spatial approach to the study of administrative linkages in the SIPP.

Data

I rely on data from the U.S. Census Bureau's Survey of Income and Program Participation (SIPP), a nationally-representative, household-based panel survey on income, employment, household composition, and government program participation that oversamples the low-income population. The 2014 panel is used for my analysis[7]; more specifically, I draw from the first wave (2014) to study initial unit nonresponse, and the second wave (2015) to study panel attrition[8]. The baseline (i.e., wave 1) sample frame included 42,348 eligible households, of which approximately 68.8% completed interviews (Treat 2017). Significant efforts were taken to track households across residential moves between survey waves; if sample members from the same household later split into separate households in a subsequent wave, all of those "spawned" households were added to the sampling frame. Households in each wave were initially contacted in-person, and interviews were conducted using face-to-face computer-assisted personal interviewing (CAPI) or, less frequently, through computer-assisted telephone interviewing (CATI) when necessary. The 2014 panel of the SIPP included an experiment on sequential incentives in which cash incentives of $0, $20, and $40 were randomly distributed to sampling units (conditional upon completion) in wave 1; in the second wave, households who participated in wave 1 were randomly assigned to a conditional incentive of either $0 or $40 (Westra et al. 2015; Ayromloo and Wilkin 2022). I account for the wave 1 incentive condition[9] in my statistical analyses.

[7] I do not include the more recently completed SIPP panel survey that began in 2018 due to changes in data collection that were implemented during the second wave (2019). As a result of these changes, the 2019 wave experienced a significant reduction in the number of respondents.
[8] Each SIPP panel contains four waves.
[9] Unfortunately, I do not have access to incentive information in wave 2.

I use SIPP paradata, which includes information from each attempt interviewers made to contact the sampling unit, as well as interviewer observations of the household and householder(s), as recorded by the interviewer in a Contact History Instrument (CHI), which is separate from the survey instrument (Bates et al. 2010; Tan 2011). I omit ineligible sample units (e.g., demolished structures, nonresidential households, households in which all members were on active duty in the armed forces), and I also omit sample units that could not be located by the interviewer[10], which represent a relatively larger share of noncontacts among panel attriters than among nonresponders in cross-sectional surveys (Lepkowski and Couper 2002; Burton et al. 2006); as mentioned previously, inter-wave geographic mobility among baseline respondents significantly contributes to panel attrition. The first wave of the 2014 panel included approximately 1,500 unique interviewers; roughly 55% of wave 1 interviewers continued to work on the second wave, which included approximately 1,200 unique interviewers. The SIPP includes a partially interpenetrated sampling design, in which interviewers were (non-randomly) assigned to households in several different counties, and several different interviewers were operating within each county. I link SIPP paradata with place-based data from the Census Bureau's Planning Database (PDB)[11] using sample units' Census tract of residence. To link sample households in the SIPP with other federal administrative records, I rely on the Census Bureau's Protected Identification Key (PIK)—i.e., a unique person identifier—assigned to each individual in the SIPP household sampling frame whose identity can be verified. I therefore use PIKs from the 2014 SIPP panel to link individuals with the Census Bureau's internal "Best Race

[10] Sampling units that could not be located had missing place-based information and thus could not contribute to estimates of place-based determinants of nonresponse.

[11] The PDB contains data from the 5-year (2013-2017) American Community Survey (ACS) and the 2010 Decennial Census. More information can be found here: www.census.gov/topics/research/guidance/planning-databases.html

and Hispanic Origin" data, which is a composite of federal, state, and third-party files (Ennis et al. 2018). Using household composition information in the SIPP, individual records through the PVS were aggregated together to generate household-level information about SIPP sampling units.

Previous analyses of sample representativeness and nonresponse bias in the 2014 SIPP panel found that, compared to responding households, nonresponders were more likely to be Black, live outside the Midwest, and reside in an urban area (Treat 2017). Also of interest is the finding that, among households that did respond to the baseline survey, participants in government programs were less likely to attrite in the second wave (Treat 2017). These findings only reflect characteristics of nonresponders for whom administrative records were available, however, which—as I will discuss in further detail—is often nonrandom. In the first wave of the 2014 SIPP panel, approximately 91% of all sampling units—and 78% of nonresponders—were successfully linked with administrative records (i.e., household race); the remaining 9% of the SIPP sample were therefore omitted from my analyses of nonresponse and attrition. To better understand the subset of the SIPP sampling frame without assigned PIKs, I examine the types of places in which they live while also accounting for the degree of spatial autocorrelation in the proportion of linked households. I draw on Census tract geospatial vector data from the National Historical GIS (NHGIS) (Manson et al. 2022).

Methods

I carry out two sets of analyses. First, to estimate the determinants of initial unit nonresponse and panel attrition broken down by noncontact and refusal, I rely on cross-classified random effects modeling (CCREM) using unweighted data. I test a separate model using

generalized linear regression for each outcome: (1) unit nonresponse due to noncontact, (2) unit nonresponse due to refusal, (3) attrition due to noncontact, and (4) attrition due to refusal; Table 1 displays sample sizes for each group. Each outcome is compared with the same group, respondents who participated in *both* waves, to facilitate comparisons across groups; as a result, sample members who participated in wave 1 but dropped out of wave 2 (i.e., attriters) were omitted from the unit nonresponse (wave 1) analysis. Because odds ratios contain unobserved heterogeneity that render comparisons across models problematic (Mood 2010), particularly with the inclusion of random effects, I estimate average marginal effects, or the average conditional effect of each variable on the probability of the outcome (Table B2, Appendix B), and apply a series of t-tests to determine whether these values differ significantly when comparing nonresponse with attrition (due to noncontact and then, separately, due to refusal). I also examine differences in the determinants of noncontact versus refusal in wave 1 (nonresponse) and, separately, in wave 2 (attrition).

My CCREM models include two random terms—one for the interviewer (at the last contact attempt), and one for the sample unit's county[12] of residence—in which sample members are separately nested. The random effects are necessarily "crossed" due to partial interpenetration in the SIPP; there are, on average, 5 interviewers per county and 3.5 counties per interviewer[13]. In their simulation of models with crossed random terms for the interviewer and area, Vassallo et al. (2017) found that having three areas per interviewer provided sufficient interpenetration. The CCREM model is ideal for this particular data structure, accounting for

[12] Counties reflect the smallest scale of geographic aggregation for which a mixed effect model of this caliber is possible; there are too many distinct Census tracts and not enough sampling units per Census tract to justify a random effect at the level of the tract.

[13] In wave 1, there were approximately 1,500 distinct interviewers and sample households were located in roughly 1,400 distinct U.S. counties.

potential similarities in survey outcomes among sample members clustered within the same county or interviewer. I draw on methods described by Nakagawa et al. (2017) to estimate intra-class correlations (ICCs) that reflect the proportion of variance in each outcome due to groupings at the level of the interviewer and county. The CCREM models also include several fixed effects, ranging from characteristics of the recruitment process, sample unit, and the sample unit's neighborhood, which I later describe in detail; see Table B1 in the supplementary appendix (Appendix B) for the results, listed as odds ratios, of all four models together. The models are specified as follows:

$$Y_{ic} = \beta_0 + \beta_1 x_{ic} + \beta_2 x_{ic} + \beta_3 x_{ic} \ldots + u_{0i} + v_{0c} + e_{ic}$$

where Y_{ic} represents one of the four outcomes (response profiles) for interviewer i and county c, and β represents a set of fixed effect coefficients for which x_{ic} is the predictor variable for interviewer i and county c. The random effects are represented by u_{0i} for the interviewer and v_{0c} for the county, and both are assumed to be normally distributed with a mean of zero; e_{ic} refers to the interviewer and county-specific error term. I estimate the proportion of variance explained by the fixed and random effects using methods described by Nakagawa and Schielzeth (2013) to calculate the pseudo R^2 in generalized linear mixed-effects models.

Second, to better understand the subset of the sampling frame for whom administrative records are unavailable, I test whether certain place characteristics are associated with the proportion of administratively linked households within a given Census tract. I undertake an explicitly spatial approach given the degree of spatial clustering in Census tract linkage rates using a queen-contiguity spatial weights matrix. The Moran's I—a commonly used measure of global spatial autocorrelation (Moran 1948)—is statistically significant ($p < 0.001$), revealing a strong degree of positive spatial autocorrelation, such that neighborhoods with similar

administrative linkage rates tend to cluster together in space. The LaGrange Multiplier diagnostic test—in its most robust form—identifies statistically significant spatial error dependence and lag dependence in the spatial distribution of neighborhood linkage rates, which, if untreated, results in biased and inefficient standard linear regression estimates (Anselin and Bera 1998; Chi and Zhu 2020), violating the underlying assumption of classic ordinary least squares (OLS) models; namely, the "independent and identically distributed" (i.i.d) assumption. I therefore employ a spatial error model with a spatially lagged response (SEMSLR), whereby the outcome—the proportion of households linked—is also treated as an independent variable in its spatially-lagged form (Chi and Zhu 2020). The model is specified as follows:

$$Y = \rho WY + \lambda WX_{lag} + \beta_0 + \beta_1 X_1 + \beta_2 X_2 \ldots + u$$

in which Y is the outcome and W is the spatial weights matrix (using queen contiguity) that captures the spatial relationships between Census tracts; ρ is the coefficient of the spatial lag of outcome Y, and WY is the spatially lagged outcome, capturing the average value of Y in neighboring Census tracts; λ is the coefficient of the spatially lagged response variable X_{lag}; β_0, $\beta_1, \beta_2 \ldots$ are the coefficients associated with the independent variables X_1, X_2, etc.; and u represents the error term is modeled endogenously as it would be in a spatial error model (capturing unobserved factors that affect the outcome and are correlated with the errors in neighboring observations). The SEMSLR regression model is also weighted using the number of sampling units within each Census tract.

Measures

I use a set of paradata, household-level, and place-based variables to predict the likelihood of four separate outcomes: (1) unit nonresponse due to noncontact, (2) unit

nonresponse due to refusal, (3) attrition due to noncontact, and (4) attrition due to refusal. Sample households with whom interviewers established contact were classified as refusals, and sample households with whom interviewers were never able to communicate (according to the CHI) were classified as "noncontacted" households. I describe the household-level predictors, as well as the predictors drawn from the paradata, below. Unless specified, the measures reflect conditions during the first wave of the panel; this facilitates comparisons of each variable across models and is also theoretically motivated by previous evidence suggesting that features of the recruitment and interview process in wave 1 is predictive of later attrition (see, for example, Pickery et al. 2001).

Contact Attempts: The total number of attempts the interviewer made to establish contact with the sampling unit, either via personal visits or outgoing telephone calls, which is typically treated as an indicator of the level of data collection effort (Williams and Brick 2018). There are a number of reasons interviewers may make several attempts to contact the sampling unit. They might, for instance, continue making personal visits to the same household until they establish contact with a householder, or they may also return to the same household more than once in order to complete the same interview in multiple stages. Across all models, I estimate the total number of contact attempts in only wave 1 to better compare the predictors of nonresponse and attrition; furthermore, separate analyses (not displayed) demonstrate that the number of contact attempts in wave 1 is more predictive of later attrition than the total number of contact attempts within the survey wave in which respondents choose to attrite. This is consistent with existing

research on the effect of prior survey experiences on later attrition (Minderop and Weiß 2022; Watson and Wooden 2009).

Outgoing Telephone Calls: Total number of times the interviewer called, as opposed to personally visited, the sampling unit, as measured in wave 1.

Number of Interviewers: Total number of unique interviewers who attempted to contact the sampling unit across all survey waves. In the nonresponse analysis, I tally the total number of unique interviewers in wave 1, and in the attrition analysis, I tally the total number of unique interviewers in waves 1 and 2 to capture the degree of interviewer continuity.

Survey-related Materials: This is a binary indicator of whether the interviewer left a voicemail, copy of a prenotice (advance) letter, contact card, or informational material with the sample household during any contact attempt within the given survey wave.

Household Concerns: Binary indicator of whether the sample member expressed any concerns about participating in the survey to the interviewer at any point during a personal visit or telephone call in wave 1. This includes concerns about time constraints, privacy, concerns "from the past," or any other expression of reluctance as observed by the interviewer.

Assigned $40 Incentive: Binary variable that indicates whether the sampling unit was randomly assigned to receive a cash incentive of $40 in wave 1; this group is compared with sampling units randomly assigned to $0 or $20 incentives. Pooling these two incentive conditions together was a methodological decision based on separate analyses in this paper as well as experimental evidence by Westra et al. (2015) and Ayromloo and Wilkin (2022) that points to the relative effectiveness of the $40 incentive compared to the other incentive conditions in reducing both unit and item nonresponse in the 2014 SIPP panel. Incentives were conditional upon survey completion in wave 1, but I only measure whether sample members were *selected* to receive the incentive, as nonresponders were unable to ever receive a post-survey incentive.

Geographic Mobility: Binary indicator of whether the sampling unit moved to a different Census tract between the first and second wave of the survey; as a result, it is only used to predict panel attrition. Previous research suggests that geographically mobile respondents are harder to contact following their move (Trappmann et al. 2015). Importantly, both Gray et al. (1996) and Lepkowski and Couper (2002) point out that, one contact is made, geographically mobile respondents are no more likely to refuse participation than respondents who remained in the same location.

Non-Hispanic White Household: Because the SIPP surveys households, rather than individuals, individual race/ethnicity data drawn from administrative records were aggregated at the level of the household to estimate a modal category. As a result, this variable represents a binary indicator of whether the majority of householders identified

as non-Hispanic White. Both households with a non-White modal category and households without a modal category were assigned a value of '0.'

Household Income. Tertile of the sample household's income as observed by the interviewer, which I treat as a continuous measure and standardize.

In addition to the variables listed above, I also include several place-based variables. Several are drawn from the PDB and reflect characteristics of the Census tracts in which sampling units reside; this includes the percentage of the population with less than a high school degree and the proportion of residents that identify as non-Hispanic White and, separately, as foreign-born. Given the importance of urbanicity in the extant literature on survey participation, I draw on Rural-Urban Commuting Area (RUCA) codes, which were assembled by the USDA Economic Research Service using ACS data, to categorize four place typologies, which include respondents who live in the (1) urban core of metropolitan statistical areas, (2) outside the urban core of metropolitan areas, (3) micropolitan statistical areas[14], and (4) small towns or rural areas. My models also estimate the effect of living in a Census-designated region (Northeast, Midwest, South, and West) considering existing evidence of regional differences in nonresponse (e.g., see Demaio 1980). It is important to note that I treat regions as an additive feature of the model, which is dissimilar to my approach in the second chapter, in which regions are interacted with the predictors of survey participation in order to account for spatial heterogeneity. (Ideally, future work would account for regional variation in the predictors of nonresponse and attrition.) Finally, I account for a binary indicator of whether households were located in group quarters

[14] Micropolitan statistical areas include a core city with 10,000 to 49,999 residents and its surrounding areas.

(e.g., university housing, correctional facilities, and nursing homes) as well as a measure of the county population size (logged).

In the spatial analysis of administrative record linkages, I test the effects of several Census tract-level characteristics drawn from the PDB: the percentage of residents between the ages of 5-17; the Gini coefficient of income inequality; and the percentage of the Census tract population (1) with less than a high school education, (2) living below the poverty line, and that (3) identifies as non-Hispanic Black and (4) foreign-born. The model also accounts for place typologies using RUCA codes as well as the four Census-designated regions.

RESULTS

Nonresponse versus Attrition: Noncontact

Table 2 displays the results of the CCREM models predicting noncontact among nonresponders (wave 1) and attriters (wave 2) as odds ratios with 95% confidence intervals (the effects of the random terms are displayed separately). The last column of Table 2 tests whether differences in the pairwise comparisons of the odds ratios for a particular variable—as measured as average marginal effects[15]—is significantly different across models. There are several notable findings. First, increases in the number of contact attempts by the interviewer is significantly associated with greater odds of noncontact, though this can arguably be described as a feature of non-contacted cases rather than a cause, in wave 1. Because this measure reflects the number of contact attempts in the *first* survey wave—for both nonresponders and attriters—this suggests that interviewers made a greater number of attempts to communicate with respondents who would later attrite (due to noncontact) than with respondents who were re-interviewed in wave 2.

[15] Average marginal effects in all four models are Table B2 (Appendix B), and the result of the four sets of significance tests are displayed in Table B3 (Appendix B).

Interestingly, however, the total number of outgoing calls by the interviewer (in wave 1) only reduces the odds of noncontact in wave 1, with no effect on noncontact in wave 2, and the difference in this effect is significant when comparing the two outcomes ($p < 0.001$). The $40 incentive (in wave 1) is only significantly associated with lower odds of noncontact in wave 1, but there is no significant difference in the effect of incentives when comparing noncontact in wave 1 with wave 2 noncontact.

There are larger differences in the household characteristics associated with non-contacted attriters versus nonresponders. Interviewers had significantly lower odds of establishing contact with households in group quarters (e.g., university housing, correctional facilities, and nursing home) in wave 1 than in wave 2. (Presumably, sample households in group quarters may be easier to contact in subsequent survey waves once contact is already established.) Household income, as estimated by the interviewer, is associated with greater odds of noncontact among both nonresponders and attriters, though the effect is significantly greater for nonresponders. There are no significant differences in the effect of household race/ethnicity (as indicated by modal non-Hispanic White households), the tract-level proportion of non-Hispanic White residents, or the tract-level proportion of foreign-born residents when comparing nonresponders with attriters. However, the proportion of Census tracts residents with less than a high school degree or GED—a measure of socioeconomic disadvantage—is associated with lower odds of noncontact in wave 1 (with no effect in wave 2, the difference of which is statistically significant). Put differently, sample members in more affluent tracts were harder to contact, but only in wave 1. In wave 1, the odds of noncontact are significantly greater in the Northeast and the South when compared to the Midwest, with no geographic variation in

noncontact in wave 2; these estimates significantly different when comparing the wave 1 and wave 2 models of noncontact.

Nonresponse versus Attrition: Refusal

Table 3 displays the predictors of refusal among nonresponders and attriters, and the last column tests whether differences in each effect is statistically significant across models. As the odds ratios in the first two columns indicate, the number of contact attempts were associated with greater odds of refusal in wave 1 but lower odds of refusal in wave 2, with a significant difference in the effect across both models ($p < 0.001$). Importantly, this variable reflects the number of contact attempts in wave 1. This suggests that, in the baseline interview, additional contact attempts is associated with the increased likelihood that a sample unit will be a nonresponder due to refusal; however, additional contact attempts that lead to participation in the first wave reduced the likelihood of attrition due to refusal in the subsequent wave (wave 2). In contrast, the number of telephone calls made by the interview appears to significantly lower the odds of refusal in wave 1 but increase the odds of refusal in the subsequent survey wave. The interpretation of this finding is also complicated but may be easier to understand by conceptualizing the analytic sample in both models as three distinct groups in the first survey wave, which consist of nonresponders, attriters, and consistent respondents. When compared with consistent respondents, nonresponders—due to refusal—received fewer telephone calls, but respondents who would later attrite (i.e., attriters due to refusal) received a greater number of telephone calls than consistent respondents; in other words, the number of times interviewers called sample members in the first wave helps predict survey refusals in entirely different ways depending on *when* the refusal occurred. The distinct number of interviewers is associated with

greater odds of refusal among both nonresponders and attriters, with a slightly greater effect on attriters ($p = 0.04$).

Assignment to a $40 incentive in wave 1 significantly reduced the odds of refusal in wave 1 and in wave 2, suggesting that the effect of participation incentives persist beyond the initial survey wave in their ability to convert potentially reluctant respondents, though it is unclear if accounting for the random incentive assignment in wave 2 would affect these results. Concerns with participation, as expressed by sample householder(s) in wave 1, were associated with greater odds of refusal in wave 1 and even in wave 2, suggesting that respondents who were reluctant to participate in the first survey wave were more likely to voluntarily drop out of the survey in the subsequent survey wave. (Household concerns in the first wave exerted a greater effect on wave 1 nonresponse than wave 2 attrition, however; $p < 0.001$.) Households in group quarters had significantly *lower* odds of refusal in wave 1, with no effect on refusals in wave 2, and the difference in the effect of group quarters is statistically significant. Compared to consistent respondents, nonresponders in wave 1—and, to a smaller degree, attriters in wave 2—were observed to have significantly higher levels of household income by interviewers. Predominantly non-Hispanic White households were also associated with significantly greater odds of refusal in both waves 1 and 2, with no significant difference in the effect across models.

There are no significant differences in the characteristics of places that predict refusal among nonresponders and attriters. Residing in a socioeconomically disadvantaged neighborhood—as indicated by the percentage of Census tract residents with less than a high school education—was associated with lower odds of refusal among both nonresponders and attriters, and, compared to residents of urban metropolitan areas, residents of small towns and rural areas were less likely to refuse participation in both wave 1 and wave 2. Notably, however,

sample members in the Northeast were more likely to refuse participation than those in the Midwest, but only in the first wave (nonresponders), with no geographic differences between attriters and respondents in wave 2. (The effect of living in the Northeast, relative to the Midwest, on cooperation is significantly different when comparing wave 1 with wave 2.)

Noncontact versus Refusal among Nonresponders (Wave 1)

Table 4 displays the predictors of noncontact and refusal among nonresponders (wave 1), testing whether the effects are significantly different in the last column. Because the separate CCREM models of nonresponse have already been covered in Tables 1 and 2, I only discuss a few notable findings in Table 4. The odds of nonresponse due to either refusal or noncontact are significantly greater as the total number of contact attempts, distinct number of interviewers, and proxy household income increase, and as the proportion of Census tract residents with less than a high school education decreases, with significant differences in the magnitude of these effects depending on the type of nonresponse. Though there are no significant racial/ethnic differences—as measured by modal non-Hispanic White household—between respondents and nonresponders whom interviewers were unable to contact, sample members in predominantly non-Hispanic White households were more likely to refuse participation in wave 1. The effect of living in a small town or rural area—compared to residents within the urban core of metropolitan areas—is only significant in predicting refusal, with no effect on noncontact; the difference in this effect across noncontact and refusal is statistically significant ($p < 0.01$). There are no significant differences in the effect of geographic regions when comparing respondents with nonresponders due to noncontact or refusal.

Noncontact versus Refusal among Attriters (Wave 2)

Finally, Table 5 displays the predictors of noncontact and refusal among attriters (wave 2) already displayed in Tables 1 and 2, with an additional column that tests whether the effects are significantly different across models. There are several results worth highlighting, beginning with the effect of the total number of contact attempts in wave 1. Compared to respondents who would go on to participate in the subsequent wave, respondents who were unable to be contacted in wave 2 required a greater number of contact attempts in wave 1 in order to complete the survey; in contrast, however, respondents who would go on to refuse participation in wave 2 required a fewer number of contact attempts to complete the survey interview when compared with consistent respondents, and the difference in the effect of contact attempts is statistically significant when comparing these two groups of attriters ($p < 0.001$). As mentioned previously, receipt of the $40 incentive in wave 1 significantly reduced refusals—but not non-contact—in wave 2, and this difference is statistically significant ($p < 0.05$). Moving between waves 1 and 2 significantly increased the odds of attrition due to both noncontact and refusal.

Certain place characteristics—including the percentage of low education and foreign-born Census tract residents, as well as the effect of residing in a small town or rural area—are only significantly associated with attrition due to refusal, and the differences in the effect of these characteristics are significant when comparing both models.

Interviewer versus Area Effects

Figure 1 displays the proportion of variance explained by both the fixed and random effects as estimated by the pseudo R^2 (Nakagawa and Schielzeth 2013) across all four CCREM models. When comparing the total variance explained by the entire model (both fixed and

random effects), the values at the top of each bar indicate that the fixed and random effects explain the greatest proportion of variance in nonresponse due to noncontact ($R^2 = 0.66$), followed by nonresponse due to refusal ($R^2 = 0.56$), attrition due to noncontact ($R^2 = 0.51$), and attrition due to refusal ($R^2 = 0.42$). The proportion of variance explained by the separate random and fixed effects appears to be patterned by nonresponse type, with the random effects (i.e., interviewer and counties) explaining a greater proportion of variance in cases of noncontact among *both* nonresponders and attriters, and the fixed effects explaining a greater proportion of variance in refusals (among nonresponders and attriters)[16].

Figure 2 offers an alternative method of evaluating the proportion of variance explained in each outcome by estimating the separate intra-class correlations (ICCs) for the interviewer and county random effects (Nakagawa et al. 2017). (Importantly, Figure 2 does not account for the effect of fixed variables in the model.) Similar to the results from the R^2 estimates (Figure 1), Figure 2 also demonstrates that the random effects, together, explain a greater proportion of variance in cases of non-contact than in cases of refusal. When breaking the ICCs down by the specific random term, however, it appears as if the greater component of variance explained by the random effects in noncontact relative to refusal is disproportionately attributable to the role of counties. Among both nonresponders and attriters due to refusal, roughly 9% of variance, on average, is explained by county-level groupings, which explain roughly 22% of variance in analyses of nonresponse and attrition due to noncontact.

Administrative Linkages

[16] In both waves, there is a greater proportion of refusals than noncontacts, which may suggest greater stability in the coefficients for the fixed effects in the models of refusal.

Given the SIPP's oversampling of low-income populations, I also investigate the extent to which linkages with administrative records are more successful among certain subsets of the sample. As mentioned previously, approximately 90% of households in the wave 1 sampling frame included at least one member with an assigned PIK. Nonresponders were significantly more "unlinkable" than respondents; on average, 78% of nonresponding households were linked with administrative records, whereas 96% of attriters were linked (put differently, 60% of unlinkable cases are nonresponders).

What do we know about the "unlinkable" members of the SIPP sampling frame? Using a spatial error model with a spatially lagged response (SEMSLR), I assess the effect of various place characteristics on neighborhood linkage rates while treating existing spatial error and lag dependence (using a queen contiguity spatial weights matrix). Even after accounting for spatial clustering in neighborhood linkage rates, the SEMLSR regression results reveal a number of neighborhood attributes associated with the proportion of linked households within a given Census tract (Table 6; see Table B4 in Appendix B for model output when spatial lag and error terms are omitted). Linkage rates significantly improve with decreases in the poverty rate and proportion of foreign-born residents. Interestingly, increases in the proportion of adolescents (ages 5 through 17) is tied to increases in Census tract linkage rates. Neither the percentage of non-Hispanic Black residents nor the percentage of those with less than a high school degree (or GED) have a significant effect on linkability, suggesting that another, omitted feature(s) of neighborhoods better predicts the linkage rate. Place typologies, which are significantly associated with linkage rates in Table B4 (Appendix B), lose their significance after treating the spatial lag and error dependence in SEMSLR regression (Table 6).

DISCUSSION

As survey response rates have continued to precipitously decline alongside recent expansions in the availability of third-party and geographic data, there has been a renewal of interest in the determinants of survey participation. Recent work has sought to identify potential sources of nonresponse that may have been overlooked in the past, resulting in separate lines of research devoted to better understanding cross-sectional unit nonresponse *or* panel attrition, with little recognition of the ways in which these two phenomena are theoretically or even empirically related.

The present study offers an extensive description of the primary categories of unit nonresponse. By drawing on the 2014 panel of the Census Bureau's Survey of Income and Program Participation (SIPP) and high-quality paradata, administrative records, and place-based data, I describe the extent to which characteristics of sample households, places, and data collection process differentially influence unit nonresponse in the first wave and panel attrition in the second wave when compared with continuous respondents in the SIPP. I further distinguish between household refusals and noncontact among nonresponders and attriters in order to more precisely identify the determinants of four major groups of unit nonresponse.

This study makes an additional methodological contribution to the literature by accounting for—and distinguishing between—the effects of interviews and places in shaping response outcomes. By exploiting the SIPP's partially interpenetrated sampling design, I use a set of cross-classified random effects models to estimate the extent to which the clustering of survey outcomes is attributable to interviewers, counties, or both. When evaluating the random effects together, I find that they explain the greatest proportion of variance among non-contacted nonresponders and attriters, and that they are almost equally attributable to both interviewers and

counties in explaining noncontact. Among nonresponders and attriters who refused to participate in the survey, interviewer groupings explain a relatively greater proportion of variance than counties. Even after including random effects for the interviewer and county, I find that increases in the number of distinct interviewers assigned to a sample unit contributed to greater refusals and noncontacts among both nonresponders and attriters, emphasizing the importance of interviewer continuity and contrasting with evidence presented by Campanelli and O'Muircheartaigh (1999).

My results also reveal an important set of differences and similarities in both the household and social determinants of nonresponse and attrition. I find that sampling units located in group quarters (e.g., university housing, correctional facilities, and nursing homes) were significantly less likely to participate in the initial survey wave of the SIPP; once they participated in the baseline interview, however, they were no more likely to attrite (for any reason) in the subsequent survey wave. Survey response behaviors also varied in terms of the places and regions in which they occurred, as well, with evidence to suggest that both nonresponders and attriters who specifically *refused* to participate were more likely to live in the urban core of metropolitan areas, rather than in small towns or rural areas, compared with continuous respondents. Interestingly, there were no differences in place typologies when comparing respondents with uncontactable nonresponders or attriters. This is a compelling finding given the prominent role of urbanicity as a social determinant of unit nonresponse, and suggests a narrower interpretation of the causal mechanisms linking place typologies with survey participation. Accounting for regional differences in each outcome also allowed me to test existing assumptions about greater survey participation and panel retention in the Midwest (Demaio 1980). Although I find that sample units in the Midwest are more likely to participate in

the initial survey wave than those in the Northeast when compared with *both* refusing and uncontactable nonresponders, there are no significant regional differences between continuous respondents and attriters, suggesting that any form of regional selection into the baseline sample does not persist into subsequent survey waves.

My findings also highlight important differences in the predictors of noncontact versus refusal, and particularly in the role of past survey experiences in predicting later attrition. When compared with respondents who would go on to participate in the next survey wave, respondents who would later refuse to participate necessitated fewer contact attempts by interviewers to complete their survey in the first wave. On the other hand, respondents who were uncontactable in the subsequent survey wave necessitated a greater number of contact attempts to complete their baseline interview compared to continuous respondents. Additionally, though both types of attriters—noncontacts and refusals—were significantly more likely to express a participation concern in the first wave when compared with continuous respondents, the expression of such a concern contributed to significantly greater refusal than noncontact among attriters. This is not entirely unsurprising, but the effect of earlier participation concerns on later attrition due to noncontact, specifically, does require further consideration. This finding may reflect the presence of an omitted variable that contributes to both participation concerns and later uncontactability. It may also reflect measurement error in interviewer documentation, as there is some evidence to suggest that interviewers underreport noncontacted cases in the Census Bureau's contact history instrument (CHI) (Bates et al. 2010).

I find that household race/ethnicity is significantly associated with panel attrition but not with initial nonresponse, similar to findings by Johnson et al. (2002); yet, the difference in the effect of household race/ethnicity across these outcomes is not statistically significant, invoking

several possible interpretations. The relationship between survey participation and race/ethnicity may need to be revisited to account for potential changes in the predictors of nonresponse or in the theoretical mechanisms linking ethnoracial identity with nonresponse. Alternatively, this finding may simply reflect weaknesses in the quality of administrative records. In a separate analysis of the characteristics of place-based linkage rates, I find no significant effect of the racial/ethnic composition, but my results do suggest that sample cases were significantly less likely to be linked with administrative records in places with relatively larger immigrant populations and populations living below the poverty line. The results of this analysis also suggest that record linkability is can be interpreted as a *spatial* phenomenon, revealing the places—and, in essence, people—that are often excluded from quantitative social science research, and highlighting the need for further investigations into the underlying attributes of places and spatial processes that contribute to this missingness.

Ultimately, researchers are interested in unit nonresponse because it serves as a potential source of nonresponse bias (Groves 2006). When the propensity to participate in a survey is systematically related to the outcome of interest, this can lead to biased inferences and limit the representativeness of the sample; relatedly, when unit nonresponse produces significant reductions in the sample size, the variances of estimates drawn from the survey data can be inflated (Dixon and Tucker 2010). Though both cross-sectional and panel surveys can be affected by this type of nonresponse bias, panel surveys are particularly vulnerable as they face the risk of baseline nonresponse bias in the initial survey wave as well as attrition bias in subsequent survey waves (Pyy-Martikainen and Rendtel 2008; Fuchs 2012; Lugtig et al. 2014). Largely due to data limitations, researchers tend to study the extent to which attrition contributes to nonresponse bias in panel surveys, with relatively less attention given to the nonresponse bias

in the initial survey wave (Watson and Wooden 2009). Underlying this approach, in many ways, is the assumption that panel attrition is simply a type of delayed nonresponse (or even "follow-up nonresponse") among respondents who were actually reluctant to participate in the prior survey wave (Cohen et al. 2000; Lugtig et al. 2014), and that the decision to participate in a survey is instead shaped by a "well-formed predisposition on the part of the sample household" that predates the initial survey request (Lepkowski and Couper 2002:262). Indeed, findings in this study reinforce the notion that many future attriters were, initially, reluctant respondents.

At the same time, my findings also highlight important gaps in the literature on nonresponse and attrition. If there is systematic selection into the initial respondent pool, as a wealth of evidence confirms (e.g., Sykes et al. 2010; Treat 2017), how does this shape downstream selection into subsequent waves of a panel survey? In Groves and Couper's seminal conceptual framework of survey participation (1998), the authors argue that the characteristics of the sampling unit, whether at the level of individual, household, or social environment, ultimately serve as approximate indicators of underlying psychosocial attributes that predict the decision to accept or refuse a survey request (p. 186). If psychosocial mechanisms indeed predict survey participation, then developing a theoretical understanding of how these mechanisms may differentially contribute to baseline nonresponse versus panel attrition may reveal deeper insight into both processes. The present study is perhaps one of the first to describe, and compare, the attributes of different response profiles in the United States, offering an empirical basis on which future research on nonresponse and attrition can build.

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TABLES

Table 1. SIPP Sample Sizes and Response Rates in Waves 1-2.

Response Profiles	N	Percent
Full Sample	44,000	100 %
Nonresponders	10,300	23.41
Continuous Respondents (Waves 1, 2)	26,500	60.23
Attriters (Responded in Wave 1)	7,200	16.40
Nonresponders	10,300	100 %
Noncontacts	1,300	12.62
Refusals	9,000	87.40
Attriters	7,200	100 %
Noncontacts	1,000	13.90
Refusals	6,200	86.11

Note. Counts represent sample households and are approximate estimates consistent with Census Bureau disclosure protocols.
Source: SIPP 2014. Census Bureau Project 6000725: CBDRB-FY23-CES010-021.

Table 2. Noncontact: Nonresponse vs. Attrition (CCREM).

	Nonresponse (W1)	Attrition (W2)	Significance Test
Contact Attempts W1	2.833*** (2.534, 3.132)	1.285*** (1.145, 1.425)	8.687 ***
Outgoing Calls W1	0.344*** (0.298, 0.390)	0.915 (0.822, 1.008)	-9.667 ***
Number of Interviewers	1.893*** (1.747, 2.039)	1.805*** (1.667, 1.943)	1.354
Received Survey Materials	4.920*** (3.492, 6.348)	3.218*** (2.567, 3.869)	1.914
Assigned $40 Incentive	0.844* (0.705, 0.984)	1.043 (0.871, 1.214)	-1.821
Household in Group Quarters	0.208* (0.057, 0.763)	1.749 (0.759, 2.739)	-3.522 ***
Household Income	2.018*** (1.819, 2.217)	1.312*** (1.240, 1.384)	6.259 ***
NH White Household	1.133 (0.933, 1.333)	1.440*** (1.176, 1.704)	-1.619
% Non-Hispanic White	1.079 (0.935, 1.223)	0.968 (0.841, 1.095)	1.149
% Foreign Born	1.068 (0.935, 1.201)	0.955 (0.838, 1.072)	1.261
% Less than HS Ed.	0.877* (0.782, 0.971)	1.089 (0.975, 1.203)	-2.838 **
County Pop Size (ln)	1.537*** (1.183, 1.891)	1.191 (0.959, 1.423)	1.697
RUCA (ref = urban Metro)			
Metro Area: Non-urban	1.141 (0.796, 1.486)	0.936 (0.657, 1.214)	0.923
Micropolitan Area	0.960 (0.572, 1.347)	0.887 (0.574, 1.200)	0.259
Small Town or Rural Area	1.438 (0.758, 2.118)	1.173 (0.718, 1.628)	0.689
Region (ref = Midwest)			
Northeast	3.180*** (1.433, 4.927)	1.350 (0.705, 1.995)	2.228 *
South	2.630*** (1.506, 3.754)	1.325 (0.850, 1.800)	2.498 *
West	1.826* (0.803, 2.849)	1.230 (0.656, 1.804)	0.942

Note. Odds ratios listed with 95% confidence intervals. Random effects are displayed separately.
Source: SIPP 2014. Census Bureau Project 6000725: CBDRB-FY23-CES010-021.
* p<0.05, ** p<0.01, *** p<0.001.

Table 3. Refusal: Nonresponse vs. Attrition (CCREM).

	Nonresponse (W1)	Attrition (W2)	Significance Test
Contact Attempts W1	1.641*** (1.554, 1.728)	0.816*** (0.770, 0.863)	17.473 ***
Outgoing Calls W1	0.719*** (0.679, 0.759)	1.139*** (1.081, 1.197)	-11.867 ***
Number of Interviewers	2.332*** (2.241, 2.423)	2.453*** (2.355, 2.551)	-2.022 *
Received Survey Materials	2.120*** (1.892, 2.348)	2.474*** (2.258, 2.690)	-2.189 *
Assigned $40 Incentive	0.831*** (0.768, 0.895)	0.920* (0.849, 0.990)	-1.806
Household Concerns W1	3.916*** (3.633, 4.199)	1.756*** (1.614, 1.898)	13.933 ***
Household in Group Quarters	0.098*** (0.019, 0.176)	0.974 (0.655, 1.292)	-6.691 ***
Household Income	1.566*** (1.480, 1.652)	1.050** (1.019, 1.081)	12.407 ***
NH White Household	1.302*** (1.192, 1.412)	1.305*** (1.194, 1.416)	-0.065
% Non-Hispanic White	1.096** (1.025, 1.167)	1.049 (0.982, 1.116)	0.930
% Foreign Born	0.998 (0.938, 1.058)	1.076* (1.011, 1.141)	-1.738
% Less than HS Ed.	0.909*** (0.863, 0.955)	0.894*** (0.848, 0.941)	0.474
County Pop Size (ln)	1.026 (0.920, 1.132)	0.955 (0.864, 1.046)	1.003
RUCA (ref = urban Metro)			
Metro Area: Non-urban	1.058 (0.918, 1.198)	1.073 (0.933, 1.213)	-0.165
Micropolitan Area	0.852 (0.705, 0.999)	0.990 (0.831, 1.149)	-1.242
Small Town or Rural Area	0.799* (0.639, 0.958)	0.798* (0.648, 0.948)	-0.006
Region (ref = Midwest)			
Northeast	1.661*** (1.200, 2.122)	1.079 (0.799, 1.359)	2.222 *
South	1.063 (0.840, 1.286)	0.984 (0.795, 1.172)	0.530
West	1.234 (0.891, 1.577)	1.089 (0.813, 1.365)	0.624

Note. Odds ratios listed with 95% confidence intervals. Random effects are displayed separately.
Source: SIPP 2014. Census Bureau Project 6000725: CBDRB-FY23-CES010-021.
* p<0.05, ** p<0.01, *** p<0.001.

Table 4. Nonresponders (Wave 1): Noncontact versus Refusal (CCREM).

	Noncontact	Refusal	Significance Test
Contact Attempts W1	2.833*** (2.534, 3.132)	1.641*** (1.554, 1.728)	-6.411 ***
Outgoing Calls W1	0.344*** (0.298, 0.390)	0.719*** (0.679, 0.759)	1.228
Number of Interviewers	1.893*** (1.747, 2.039)	2.332*** (2.241, 2.423)	-27.677 ***
Received Survey Materials	4.920*** (3.492, 6.348)	2.120*** (1.892, 2.348)	-7.154 ***
Assigned $40 Incentive	0.844* (0.705, 0.984)	0.831*** (0.768, 0.895)	3.170 **
Household in Group Quarters	0.208* (0.057, 0.763)	0.098*** (0.019, 0.176)	7.616 ***
Household Income	2.018*** (1.819, 2.217)	1.566*** (1.480, 1.652)	-7.89 ***
NH White Household	1.133 (0.933, 1.333)	1.302*** (1.192, 1.412)	-4.714 ***
% Non-Hispanic White	1.079 (0.935, 1.223)	1.096** (1.025, 1.167)	-1.899
% Foreign Born	1.068 (0.935, 1.201)	0.998 (0.938, 1.058)	0.553
% Less than HS Ed.	0.877* (0.782, 0.971)	0.909*** (0.863, 0.955)	2.017 *
County Pop Size (ln)	1.537*** (1.183, 1.891)	1.026 (0.920, 1.132)	1.405
RUCA (ref = urban Metro)			
Metro Area: Non-urban	1.141 (0.796, 1.486)	1.058 (0.918, 1.198)	-0.265
Micropolitan Area	0.960 (0.572, 1.347)	0.852 (0.705, 0.999)	1.468
Small Town or Rural Area	1.438 (0.758, 2.118)	0.799* (0.639, 0.958)	2.622 **
Region (ref = Midwest)			
Northeast	3.180*** (1.433, 4.927)	1.661*** (1.200, 2.122)	-1.430
South	2.630*** (1.506, 3.754)	1.063 (0.840, 1.286)	1.445
West	1.826* (0.803, 2.849)	1.234 (0.891, 1.577)	-0.555

Note. Odds ratios listed with 95% confidence intervals. Random effects are displayed separately.
Source: SIPP 2014. Census Bureau Project 6000725: CBDRB-FY23-CES010-021.
* p<0.05, ** p<0.01, *** p<0.001.

Table 5. Attriters (Wave 2): Noncontact versus Refusal (CCREM).

	Noncontact	Refusal	Significance Test
Contact Attempts W1	1.285*** (1.145, 1.425)	0.816*** (0.770, 0.863)	8.157 ***
Outgoing Calls W1	0.915 (0.822, 1.008)	1.139*** (1.081, 1.197)	-5.234 ***
Number of Interviewers	1.805*** (1.667, 1.943)	2.453*** (2.355, 2.551)	-28.071 ***
Received Survey Materials	3.218*** (2.567, 3.869)	2.474*** (2.258, 2.690)	-12.953 ***
Assigned $40 Incentive	1.043 (0.871, 1.214)	0.920* (0.849, 0.990)	2.148 *
Household Concerns W1	1.270** (1.043, 1.497)	1.756*** (1.614, 1.898)	-10.16 ***
Household Moved	3.508*** (2.390, 4.626)	1.327** (1.059, 1.595)	0.152
Household in Group Quarters	1.749 (0.759, 2.739)	0.974 (0.655, 1.292)	0.980
Household Income	1.312*** (1.240, 1.384)	1.050** (1.019, 1.081)	0.914
NH White Household	1.440*** (1.176, 1.704)	1.305*** (1.194, 1.416)	-3.759 ***
% Non-Hispanic White	0.968 (0.841, 1.095)	1.049 (0.982, 1.116)	-1.524
% Foreign Born	0.955 (0.838, 1.072)	1.076* (1.011, 1.141)	-2.467 *
% Less than HS Ed.	1.089 (0.975, 1.203)	0.894*** (0.848, 0.941)	4.483 ***
County Pop Size (ln)	1.191 (0.959, 1.423)	0.955 (0.864, 1.046)	1.613
RUCA (ref = urban Metro)			
Metro Area: Non-urban	0.936 (0.657, 1.214)	1.073 (0.933, 1.213)	-1.143
Micropolitan Area	0.887 (0.574, 1.200)	0.990 (0.831, 1.149)	-0.195
Small Town or Rural Area	1.173 (0.718, 1.628)	0.798* (0.648, 0.948)	2.486 *
Region (ref = Midwest)			
Northeast	1.350 (0.705, 1.995)	1.079 (0.799, 1.359)	-0.055
South	1.325 (0.850, 1.800)	0.984 (0.795, 1.172)	0.7660
West	1.230 (0.656, 1.804)	1.089 (0.813, 1.365)	-0.288

Note. Odds ratios listed with 95% confidence intervals. Random effects are displayed separately.
Source: SIPP 2014. Census Bureau Project 6000725: CBDRB-FY23-CES010-021.
* p<0.05, ** p<0.01, *** p<0.001.

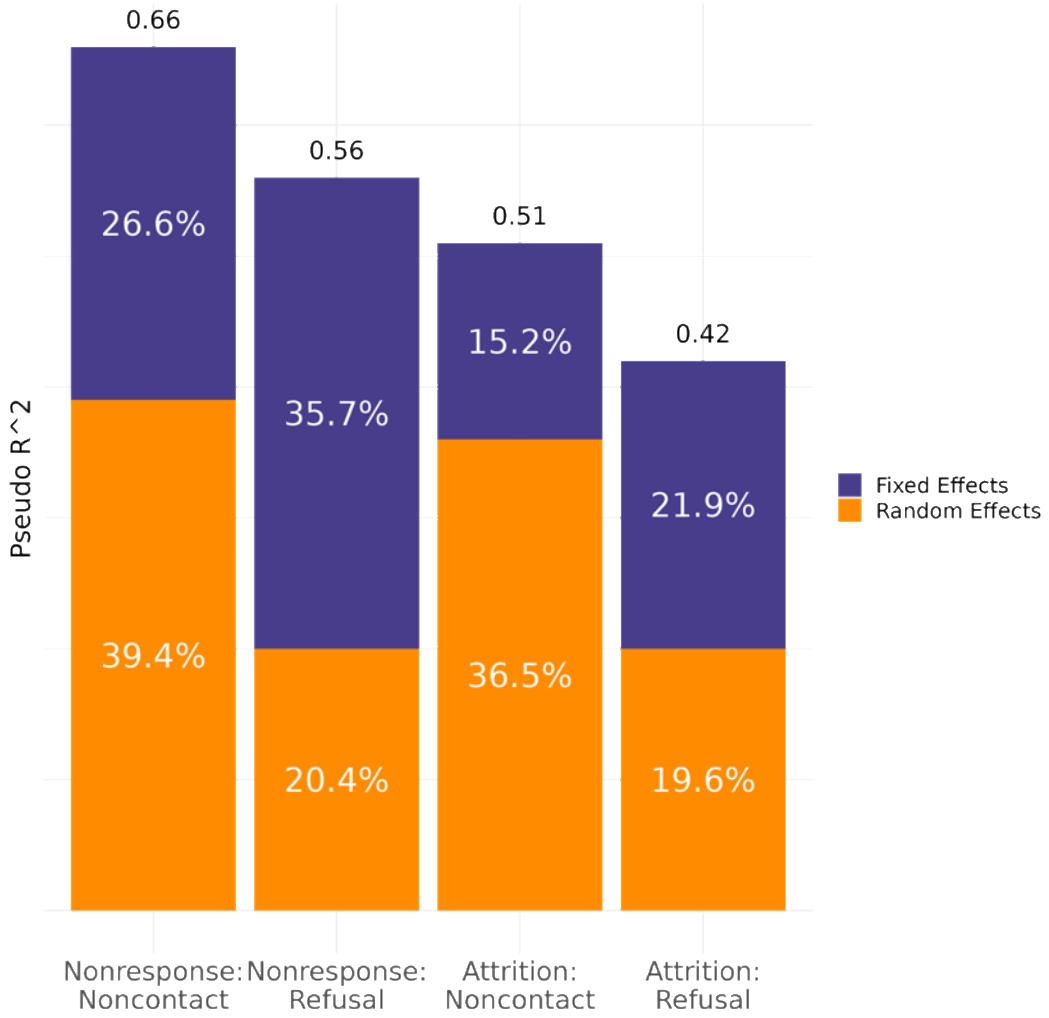
Table 6. SEMSLR: Percentage of Administratively Linked Cases per Census Tract.

	Coefficients	95% CI
% Age 5-17	0.097 ***	(0.057, 0.136)
% Below Poverty	-0.045 ***	(-0.067, -0.023)
Gini Index	0.027	(-0.027, 0.080)
% Non-Hispanic Black	0.005	(-0.004, 0.014)
% Foreign-born	-0.039 ***	(-0.058, -0.021)
% Less than HS Ed.	0.015	(-0.010, 0.039)
RUCA (ref = urban Metro)		
Metro Area: Non-urban	-0.192	(-0.833, 0.449)
Micropolitan Area	0.183	(-0.418, 0.784)
Small Town or Rural Area	-0.125	(-0.894, 0.644)
Region (ref = Midwest)		
Northeast	-0.122	(-0.640, 0.396)
South	-0.018	(-0.418, 0.383)
West	0.139	(-0.335, 0.613)
Spatial Lag	0.906 ***	(0.896, 0.916)
Spatial Error (lambda)	-0.77 ***	(-0.777, -0.762)

Note. The spatial lag term is a spatially-lagged version of the outcome (percentage of linked cases per Census tract). The model includes weights for the number of sample cases per tract, and also adjusts for the logged county population size (not displayed).
Source: SIPP 2014. Census Bureau Project 6000725: CBDRB-FY23-CES010-021.
* p<0.05, ** p<0.01, *** p<0.001.

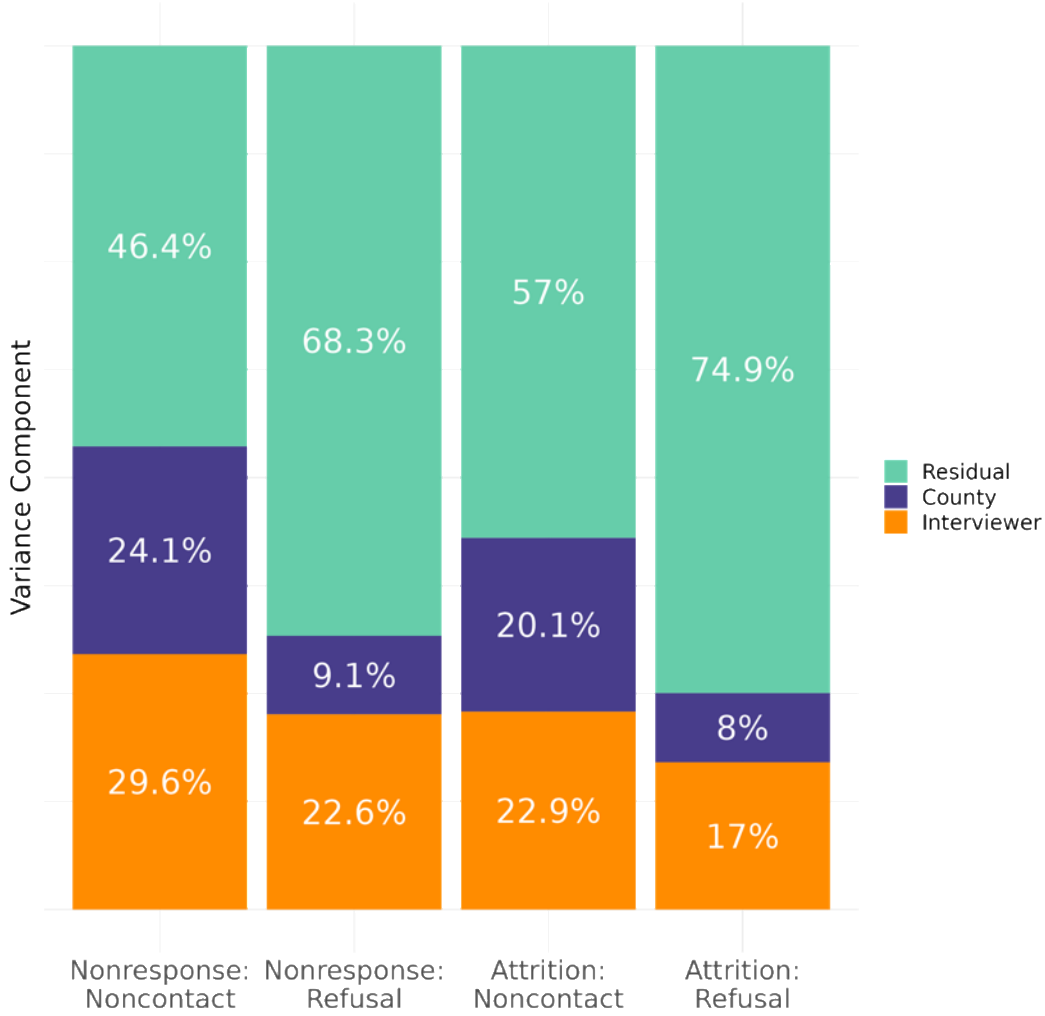
FIGURES

Figure 1. Proportion of Variance Explained by Fixed and Random Effects (Pseudo R^2).



Source: SIPP 2014. Census Bureau Project 6000725: CBDRB-FY23-CES010-021.

Figure 2. Proportion of Interviewer, County, and Residual Variance (ICC).



Note. Variance components estimated using intra-class correlations (ICCs).
Source: SIPP 2014. Census Bureau Project 6000725: CBDRB-FY23-CES010-021.

APPENDIX B

Table B1. Odds Ratios

	Nonresponse		Attrition	
	(1) Noncontact	(2) Refusal	(3) Noncontact	(4) Refusal
Contact Attempts W1	2.833*** (2.534, 3.132)	1.641*** (1.554, 1.728)	1.285*** (1.145, 1.425)	0.816*** (0.770, 0.863)
Outgoing Calls W1	0.344*** (0.298, 0.390)	0.719*** (0.679, 0.759)	0.915 (0.822, 1.008)	1.139*** (1.081, 1.197)
Number of Interviewers	1.893*** (1.747, 2.039)	2.332*** (2.241, 2.423)	1.805*** (1.667, 1.943)	2.453*** (2.355, 2.551)
Received Survey Materials	4.920*** (3.492, 6.348)	2.120*** (1.892, 2.348)	3.218*** (2.567, 3.869)	2.474*** (2.258, 2.690)
Assigned $40 Incentive	0.844* (0.705, 0.984)	0.831*** (0.768, 0.895)	1.043 (0.871, 1.214)	0.920* (0.849, 0.990)
Household in Group Quarters	0.208* (0.057, 0.763)	0.098*** (0.019, 0.176)	1.749 (0.759, 2.739)	0.974 (0.655, 1.292)
Household Income	2.018*** (1.819, 2.217)	1.566*** (1.480, 1.652)	1.312*** (1.240, 1.384)	1.050** (1.019, 1.081)
NH White Household	1.133 (0.933, 1.333)	1.302*** (1.192, 1.412)	1.440*** (1.176, 1.704)	1.305*** (1.194, 1.416)
% Non-Hispanic White	1.079 (0.935, 1.223)	1.096** (1.025, 1.167)	0.968 (0.841, 1.095)	1.049 (0.982, 1.116)
% Foreign Born	1.068 (0.935, 1.201)	0.998 (0.938, 1.058)	0.955 (0.838, 1.072)	1.076* (1.011, 1.141)
% Less than HS Ed.	0.877* (0.782, 0.971)	0.909*** (0.863, 0.955)	1.089 (0.975, 1.203)	0.894*** (0.848, 0.941)
County Pop Size (ln)	1.537*** (1.183, 1.891)	1.026 (0.920, 1.132)	1.191 (0.959, 1.423)	0.955 (0.864, 1.046)
RUCA (ref = urban Metro)				
Metro Area: Non-urban	1.141 (0.796, 1.486)	1.058 (0.918, 1.198)	0.936 (0.657, 1.214)	1.073 (0.933, 1.213)
Micropolitan Area	0.960 (0.572, 1.347)	0.852 (0.705, 0.999)	0.887 (0.574, 1.200)	0.990 (0.831, 1.149)
Small Town or Rural Area	1.438 (0.758, 2.118)	0.799* (0.639, 0.958)	1.173 (0.718, 1.628)	0.798* (0.648, 0.948)
Region (ref = Midwest)				
Northeast	3.180*** (1.433, 4.927)	1.661*** (1.200, 2.122)	1.350 (0.705, 1.995)	1.079 (0.799, 1.359)
South	2.630*** (1.506, 3.754)	1.063 (0.840, 1.286)	1.325 (0.850, 1.800)	0.984 (0.795, 1.172)
West	1.826* (0.803, 2.849)	1.234 (0.891, 1.577)	1.230 (0.656, 1.804)	1.089 (0.813, 1.365)
Household Concerns W1		3.916*** (3.633, 4.199)	1.270** (1.043, 1.497)	1.756*** (1.614, 1.898)
Household Moved			3.508*** (2.390, 4.626)	1.327** (1.059, 1.595)

Source: SIPP 2014. Census Bureau Project 6000725: CBDRB-FY23-CES010-021.
* p<0.05, ** p<0.01, *** p<0.001.

Table B2. Average Marginal Effects with 95% CIs.

	Nonresponse (W1)		Attrition (W2)	
	Noncontact	Refusal	Noncontact	Refusal
Contact Attempts W1	0.032 ***	0.057 ***	0.007 ***	-0.023 ***
Outgoing Calls W1	-0.032 ***	-0.038 ***	-0.002	0.015 ***
Number of Interviewers	0.019 ***	0.097 ***	0.016 ***	0.104 ***
Received Survey Materials	0.035 ***	0.081 ***	0.027 ***	0.097 ***
Assigned $40 Incentive	-0.005 *	-0.021 ***	0.001	-0.010 *
Household in Group Quarters	-0.03 ***	-0.179 ***	0.019	-0.003
Household Income	0.021 ***	0.051 ***	0.007 ***	0.006 **
NH White Household	0.004	0.03 ***	0.010 ***	0.030 ***
% Non-Hispanic White	0.002	0.01 **	-0.001	0.006
% Foreign Born	0.002	0.000	-0.001	0.008 *
% Less than HS Ed.	-0.004 *	-0.011 ***	0.002	-0.013 ***
County Pop Size (ln)	0.013 ***	0.003	0.005	-0.005
RUCA (ref = urban Metro)				
Metro Area: Non-urban	0.004	0.007	-0.002	0.008
Micropolitan Area	-0.001	-0.018	-0.003	-0.001
Small Town or Rural Area	0.012	-0.025 *	0.005	-0.025 *
Region (ref = Midwest)				
Northeast	0.033 ***	0.06 ***	0.008	0.009
South	0.026 ***	0.007	0.007	-0.002
West	0.014 *	0.024	0.005	0.010
Household Concerns W1		0.181 ***	0.007 *	0.07 ***
Household Moved			0.035 ***	0.033 **

Source: SIPP 2014. Census Bureau Project 6000725: CBDRB-FY23-CES010-021.
* p<0.05, ** p<0.01, *** p<0.001.

Table B3. Tests of Significance across all CCREM Models.

	Nonresponse vs. Attrition		Refusal vs. Noncontact	
	(1) Noncontact	(2) Refusal	(3) Nonresponse	(4) Attrition
Contact Attempts W1	8.687 ***	17.473 ***	-6.411 ***	8.157 ***
Outgoing Calls W1	-9.667 ***	-11.867 ***	1.228	-5.234 ***
Number of Interviewers	1.354	-2.022 *	-27.677 ***	-28.071 ***
Received Survey Materials	1.914	-2.189 *	-7.154 ***	-12.953 ***
Assigned $40 Incentive	-1.821	-1.806	3.170 **	2.148 *
Household in Group Quarters	-3.522 ***	-6.691 ***	7.616 ***	0.980
Household Income	6.259 ***	12.407 ***	-7.89 ***	0.914
NH White Household	-1.619	-0.065	-4.714 ***	-3.759 ***
% Non-Hispanic White	1.149	0.930	-1.899	-1.524
% Foreign Born	1.261	-1.738	0.553	-2.467 *
% Less than HS Ed.	-2.838 **	0.474	2.017 *	4.483 ***
County Pop Size (ln)	1.697	1.003	1.405	1.613
RUCA (ref = urban Metro)				
Metro Area: Non-urban	0.923	-0.165	-0.265	-1.143
Micropolitan Area	0.259	-1.242	1.468	-0.195
Small Town or Rural Area	0.689	-0.006	2.622 **	2.486 *
Region (ref = Midwest)				
Northeast	2.228 *	2.222 *	-1.430	-0.055
South	2.498 *	0.530	1.445	0.7660
West	0.942	0.624	-0.555	-0.288
Household Concerns W1	-	13.933 ***	-	-10.16 ***
Household Moved	-	-	-	0.152

Source: SIPP 2014. Census Bureau Project 6000725: CBDRB-FY23-CES010-021.
* p<0.05, ** p<0.01, *** p<0.001.

Table B4. Non-spatial model (OLS) of Census tract record linkages.

	Coefficients	Standard Errors
% Age 5-17	0.335 ***	(0.03)
% Below Poverty	-0.169 ***	(0.02)
Gini Index	-0.196 ***	(0.06)
% Non-Hispanic Black	-0.005	(0.01)
% Foreign-born	-0.249 ***	(0.02)
% Less than HS Ed.	0.058	(0.02)
RUCA (ref = urban Metro)		
Metro Area: Non-urban	-0.764	(0.59)
Micropolitan Area	0.001	(0.63)
Small Town or Rural Area	-2.030 **	(0.75)
Region (ref = Midwest)		
Northeast	-1.218 *	(0.57)
South	-0.540	(0.44)
West	-0.851	(0.52)
Spatial Lag	-	-
Spatial Error (lambda)	-	-

Note. The model includes weights for the number of sample cases per tract, and adjusts for the logged county population size (not displayed).
Source: SIPP 2014. Census Bureau Project 6000725: CBDRB-FY23-CES010-021.
* p<0.05, ** p<0.01, *** p<0.001.